Task 2 report

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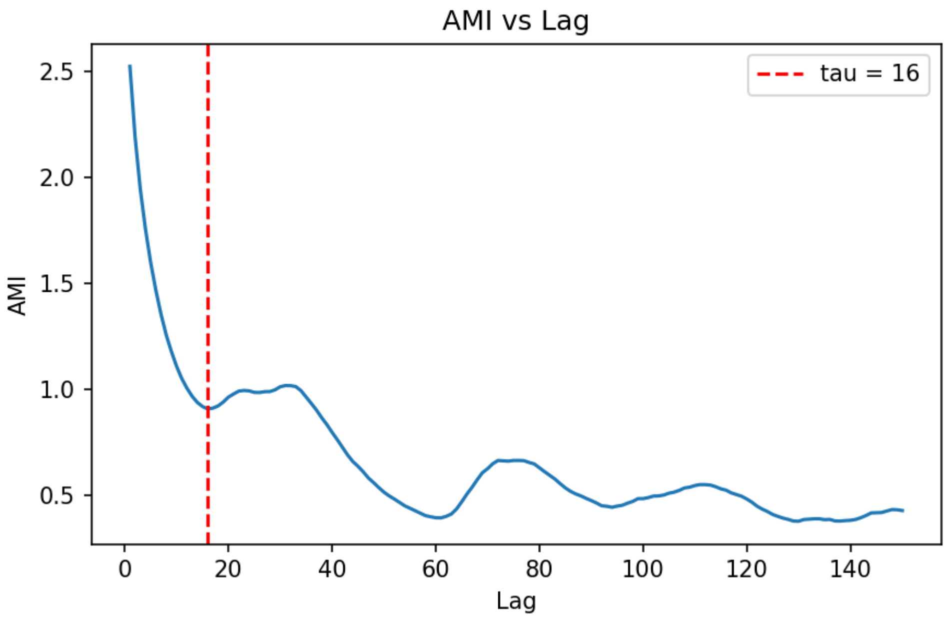
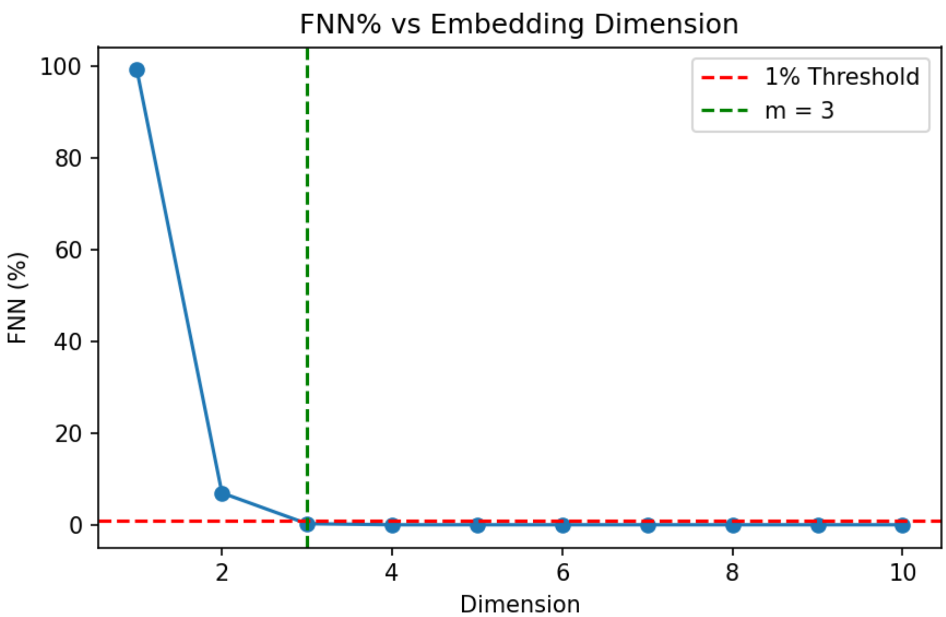
Part A: Phase Space Reconstruction

The first step is reconstructing the state space from a single scalar variable x(t). We calculated the optimal time delay using Average Mutual Information (AMI). The first local minimum gives tau = 16, representing the optimal lag for independence while preserving dynamics.



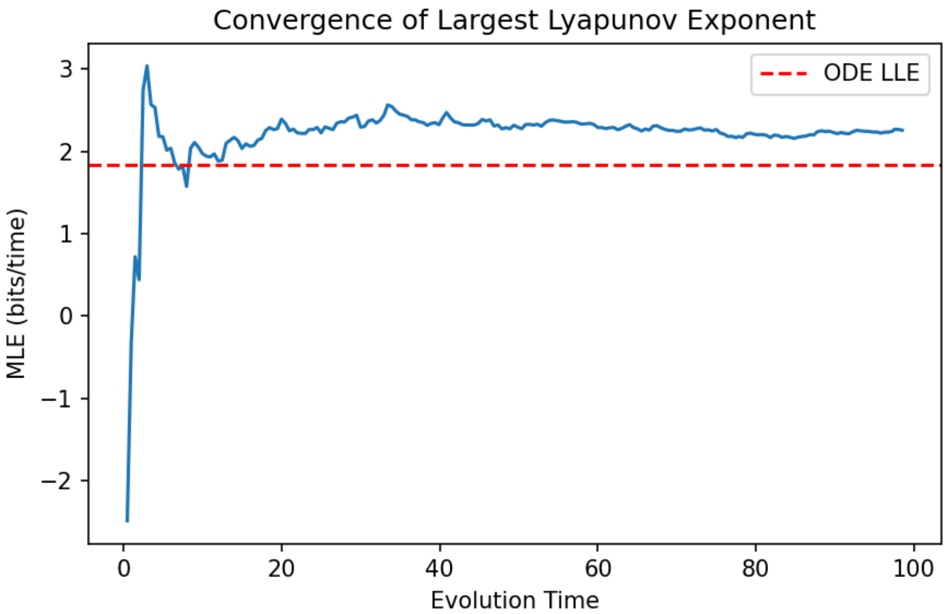
Using tau = 16, we then determined the optimal embedding dimension using the False Nearest Neighbors (FNN) percentage. When the dimension reached m = 3, the FNN dropped below 1%, confirming 3 is the ideal dimension for topological reconstruction. This optimally matches the known original state dimension of the Lorenz system, which is 3.



Part B: Lyapunov Exponent

Lyapunov exponents quantify the rate of separation of infinitesimally close trajectories. Using Wolf's Gram-Schmidt method on the ODE directly, we found the full Lyapunov spectrum.

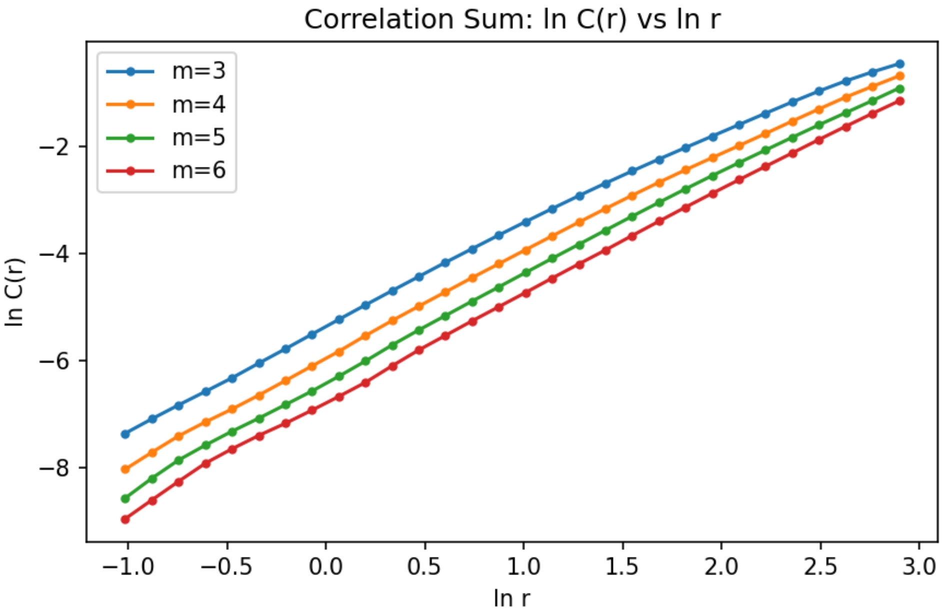
We then estimated the Largest Lyapunov Exponent (LLE) strictly from the reconstructed time-series data using Wolf's Phase-Space tracking. The time-series MLE result was 2.2561 bits/time (with 200 segments tracked).



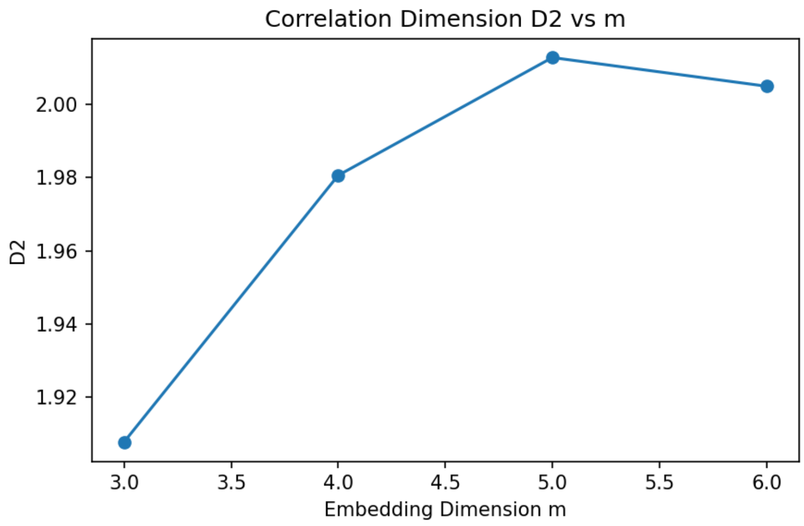
Both methods indicate one positive Lyapunov exponent, effectively classifying the Lorenz system as chaotic. The slight discrepancy in the value is expected due to the sensitivity of tracking parameters and finite series length.

Part C & D: Correlation Dimension & K2 Entropy

We calculated the Correlation Sum C(r) for multiple embedding dimensions and plotted its logarithm.



The Correlation Dimension D2 is the slope of the linear scaling region. Plotting D2 against embedding dimension shows that it saturates at approximately 2.00. This non-integer, saturated dimension is a strong indicator of low-dimensional chaos rather than random noise.

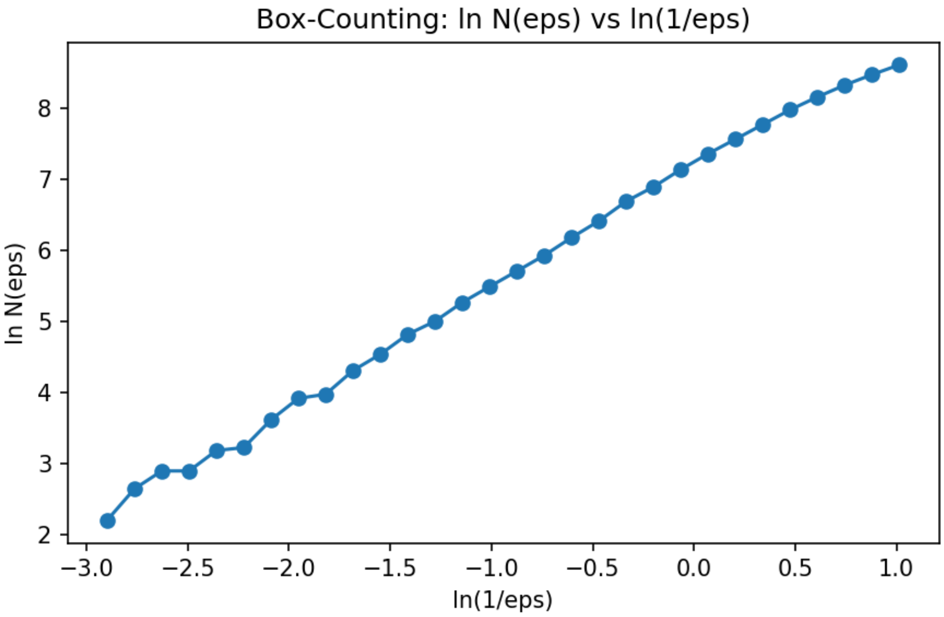


The K2 Entropy (Kolmogorov-Sinai) estimates the rate of information production. By taking the ratio of correlation sums between consecutive dimensions, our K2 estimates saturate to finite, positive values:

K2 values: 54.27, 43.62, 36.69

Part E: Fractal (Box-Counting) Dimension

The Box-Counting Dimension (D0) measures the topological dimension by covering the attractor with grids of size eps. The slope of ln N(eps) against ln(1/eps) yields the fractal dimension.



We calculated D0 to be approximately 1.77. As expected for fractal chaotic attractors, D0 (1.77) is greater than or roughly equal to the Correlation Dimension D2 (2.00).

Conclusion

This report quantitatively evaluated the Lorenz system. Through Average Mutual Information and False Nearest Neighbors, we successfully reconstructed the phase space. The positive Largest Lyapunov exponent confirmed its chaotic nature. Finally, the saturated fractional dimensions (Correlation D2 and Box-counting D0) alongside finite positive K2 entropy estimates robustly established that the underlying dynamics represent a low-dimensional chaotic strange attractor.